

MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

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Extended Abstract—Breast cancer is the most commonly diagnosed cancer worldwide. Population screening with full-field digital mammography (FFDM) reduces mortality but places a heavy radiologist workload and substantial inter-reader variability of 10–30 percentage points (Lehman et al., 2017). Meta-analyses report pooled AUC of 0.87 for deep learning AI versus 0.81 for radiologists on digital mammography (Dembrower et al., 2020). Three gaps motivate this work: single-view dominance in published VinDr-Mammo results, underexplored spatial cross-view attention, and the absence of uncertainty-aware triage simulation on this dataset.

We developed MamNet-6, a six-stage pipeline evaluated on VinDr-Mammo (Nguyen et al., 2023), a 20,000-image Vietnamese FFDM benchmark. Each stage extends the previous: Stage 1 establishes CNN baselines across five backbones; Stage 2 introduces BreastPairNet, a shared-backbone multi-view network fusing CC and MLO views across twelve backbone $\times$ fusion $\times$ resolution variants; Stage 3 adds a CrossView Transformer with spatial cross-attention between view feature maps before global average pooling; Stage 3b tests whether coarse bounding-box annotations help or hurt performance; Stage 4 applies Monte Carlo (MC) Dropout ($N=20$ stochastic passes) for predictive uncertainty quantification; and Stage 5 assembles a 16-model ensemble with test-time augmentation (TTA $\times 5$). Stage 6 simulates a Verboom-style hybrid triage policy (Verboom et al., 2021), routing high-confidence AI reads autonomously and referring uncertain cases to a radiologist. External zero-shot validation is performed on the RSNA 2022 dataset.

DenseNet-121 achieved AUROC = 0.756 at Stage 1. BreastPairNet improved to AUROC = 0.826 [95 % CI 0.776–0.871] (+9.1 %), and the CrossView Transformer reached AUROC = 0.832 [0.783–0.875] (+10.0 %). Adjacent-stage differences lie within overlapping confidence intervals; the cumulative Stage 1 $\rightarrow$ 5 gain is the most statistically robust comparison. Bounding-box annotations consistently degraded AUROC by 0.12–0.20, indicating shortcut learning from coarse localisation labels. The 16-model ensemble achieved AUROC = 0.846 (+11.8 % vs. Stage 1) with sensitivity at 90 % specificity ($S@90\%spec$) = 0.657 and positive predictive value (PPV) = 0.184 at the triage threshold. Hybrid triage at 50 % AI reads preserved 87.9 % sensitivity under oracle conditions. Zero-shot transfer to RSNA 2022 yielded AUROC = 0.601, highlighting a large domain gap between Vietnamese and North American screening populations.

MamNet-6 contributes a systematic ablation across six design decisions—backbone, multi-view fusion, spatial cross-attention, uncertainty quantification, ensemble construction, and clinical triage simulation—on a single large FFDM benchmark, delivering a cumulative 11.8 % AUROC gain over single-view CNN. Two key negative findings emerge: bounding-box shortcut learning degrades performance, and domain generalisation to unseen screening populations remains an open challenge.

References

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